

Short-Term Rainfall Prediction Applied to an Urban Basin

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Urban flooding in rapidly responding urban basins represents a major challenge for hydrometeorological monitoring systems due to the occurrence of short-duration convective precipitation events [1]. This work presents a machine learning approach for short-term rainfall prediction using pluviometric time series collected from automatic rainfall stations distributed across an urban basin [2, 3].

A multistation temporal dataset with 10-minute resolution was constructed from rainfall measurements collected from five monitoring stations located in Asunción and its metropolitan area. An automatic rainfall event detection methodology was implemented using a minimum precipitation threshold of 0.3 mm/10 min and maximum dry intervals between consecutive records. For each detected event, temporal accumulation features were extracted from past rainfall measurements obtained between 10 and 30 minutes before the event using data from both local and neighboring stations.

Linear Regression, Random Forest, and XGBoost models were evaluated for predicting three target variables: rainfall peak intensity, event duration, and future 30-minute rainfall accumulation. Results showed that tree-based models outperformed linear regression approaches for the three target variables, indicating nonlinear relationships in rainfall dynamics. The best performance for accumulated rainfall prediction was obtained using a multistation XGBoost model, achieving coefficients of determination close to $R^2 \approx 0.56$. For rainfall peak prediction, Random Forest models achieved coefficients close to $R^2 \approx 0.73$. Figure 1 presents the relationship between observed and predicted 30-minute rainfall accumulation obtained using the multistation XGBoost model.

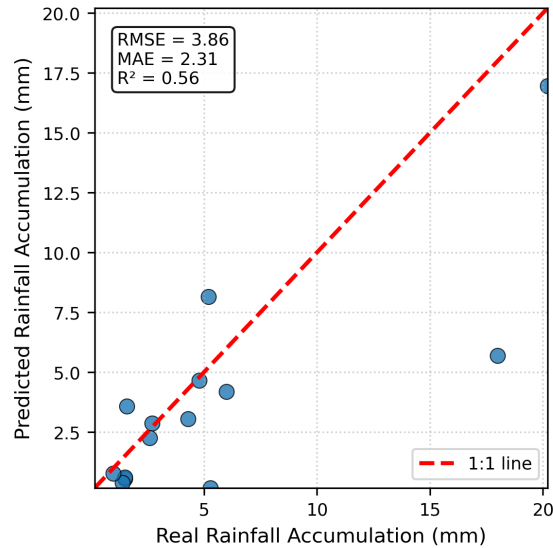


Figure 1: Real versus predicted 30-minute rainfall accumulation using the multistation XGBoost model.

The results suggest that multistation rainfall data can improve short-term rainfall prediction in urban basins. Future work will include the incorporation of additional monitoring stations and satellite-derived products in order to improve model performance.

References

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